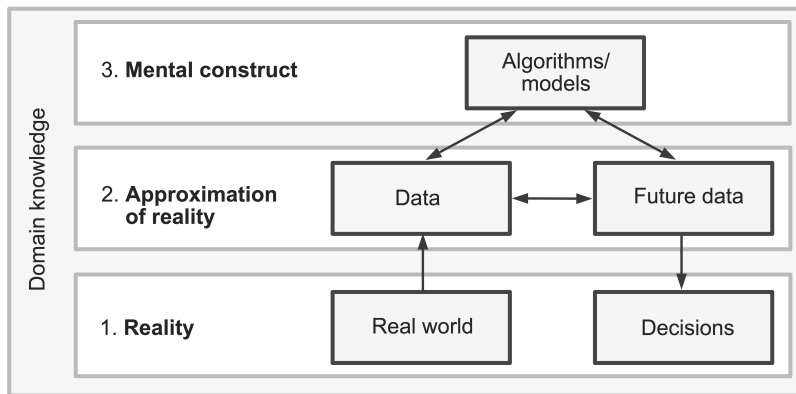


Assessing predictions

Ciaran Evans

Model assessment

Essentially, all models are wrong, but some are useful. (Box and Draper, 1987)



Today: Is our good at *predicting* the response?

Warmup

Work on the warmup activity (handout).

Types of prediction

Regression:

Classification:

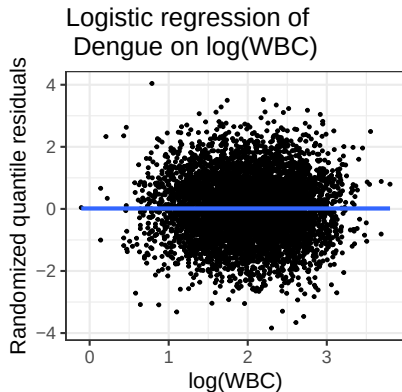
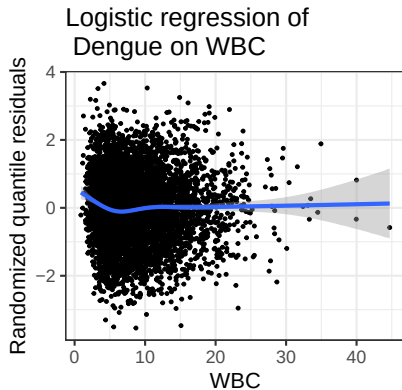
General principle: predictions from regression can be converted into classification, but not the other way around!

Assessing predicted probabilities

$$Y_i | \mathbf{x}_i \sim \text{Bernoulli}(p_i)$$
$$\log \left(\frac{\hat{p}_i}{1 - \hat{p}_i} \right) = \mathbf{x}_i^T \hat{\boldsymbol{\beta}}$$

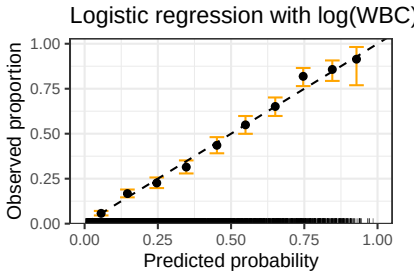
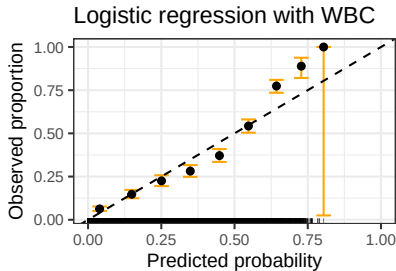
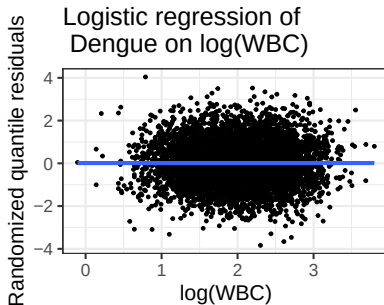
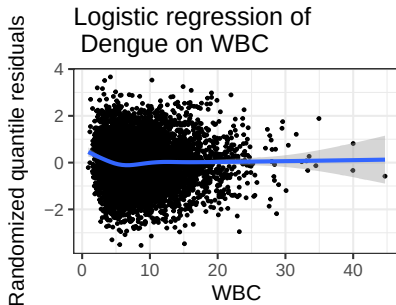
Question: How do we tell if $\hat{p}_i$ is a “good” estimator of p_i ?

Assessing predicted probabilities



Question: Does using WBC vs. $\log(\text{WBC})$ make a difference for the quality of our predictions?

Calibration plot



Assessing binary predictions

Question: Logistic regression gives me $\hat{p}_i$. How do I turn $\hat{p}_i \in (0, 1)$ into $\hat{Y}_i \in \{0, 1\}$?

Confusion matrix

	$Y = 0$	$Y = 1$
$\hat{Y} = 0$	True Negatives (TN)	False Negatives (FN)
$\hat{Y} = 1$	False Positives (FP)	True Positives (TP)

Confusion matrix

	No active disease	Active disease
FC < 61 mcg/g	15	7
FC $\geq$ 61 mcg/g	3	37